US 20210101818A1

(19) **United States**

(12) **Patent Application Publication**

Drewnowski et al.

(10) **Pub. No.: US 2021/0101818 A1**

(43) **Pub. Date: Apr. 8, 2021**

(54) **ADDITIVE MANUFACTURING SYSTEMS AND METHOD FOR MAKING GLASS ARTICLES**

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(21) Appl. No.: **16/464,563**

(22) PCT Filed: **Nov. 27, 2017**

(86) PCT No.: **PCT/US2017/063287**

§ 371 (c)(1),
(2) Date: **May 28, 2019**

Related U.S. Application Data

(60) Provisional application No. 62/426,895, filed on Nov. 28, 2016.

Publication Classification

(51) **Int. Cl.**

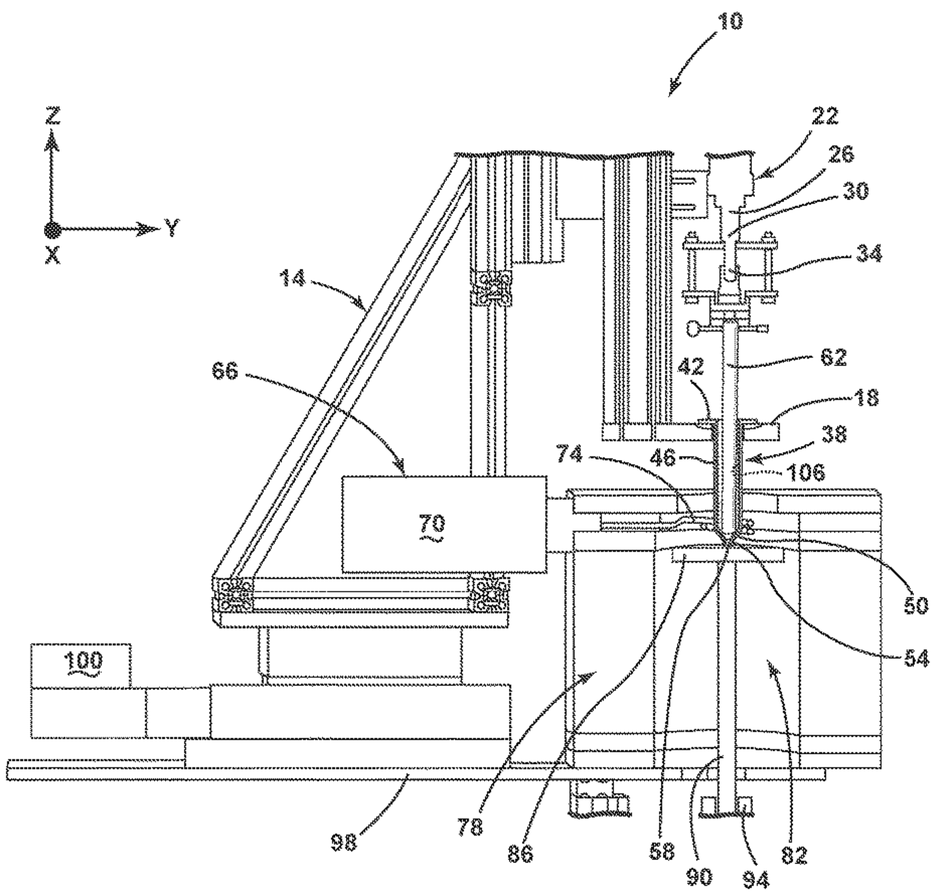
C03B 19/02	(2006.01)
C03B 5/02	(2006.01)
C03B 25/00	(2006.01)

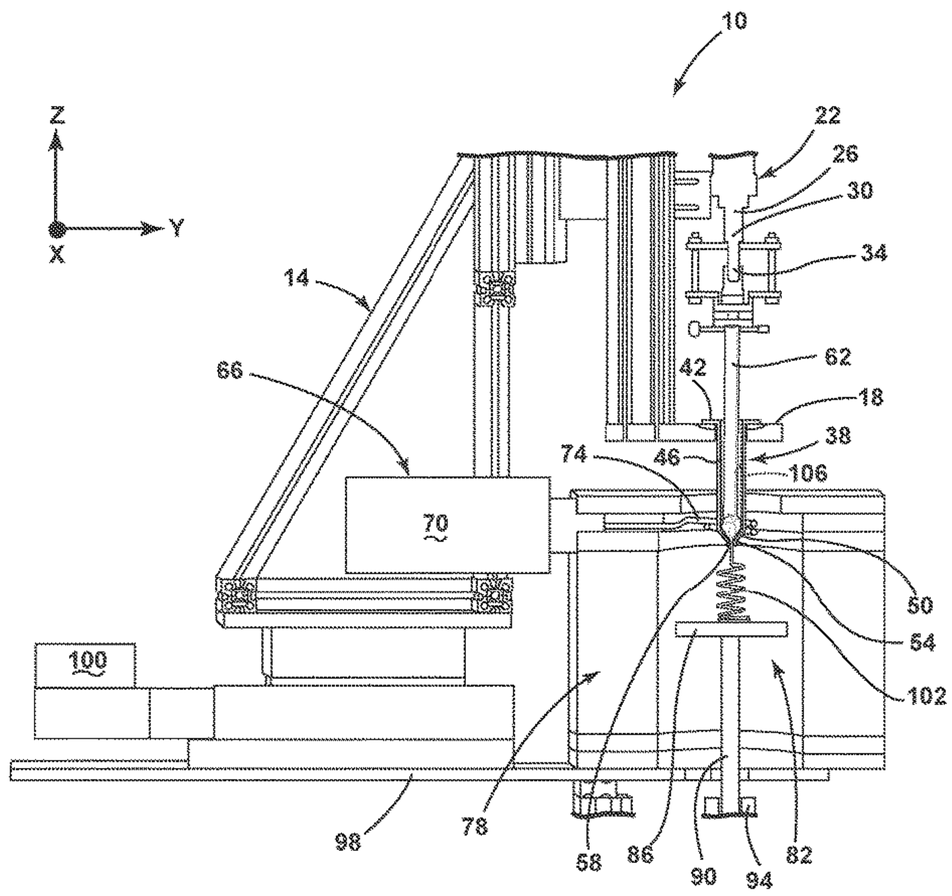
(52) **U.S. Cl.**

CPC ***C03B 19/025*** (2013.01); *B33Y 10/00* (2014.12); ***C03B 25/00*** (2013.01); ***C03B 5/021*** (2013.01)

(57) **ABSTRACT**

A glass article manufacturing system (**10**) includes a crucible (**38**) that defines a barrel (**46**) and a nozzle (**54**). The barrel (**46**) accepts a glass feedstock (**62**). A heater **66** is in thermal communication with the nozzle (**54**). The heater **66** heats the feedstock (**62**) within the nozzle (**54**). An actuator (**22**) is positioned proximate the barrel (**46**) and extrudes the feedstock (**62**) through the nozzle (**54**) as extruded feedstock.





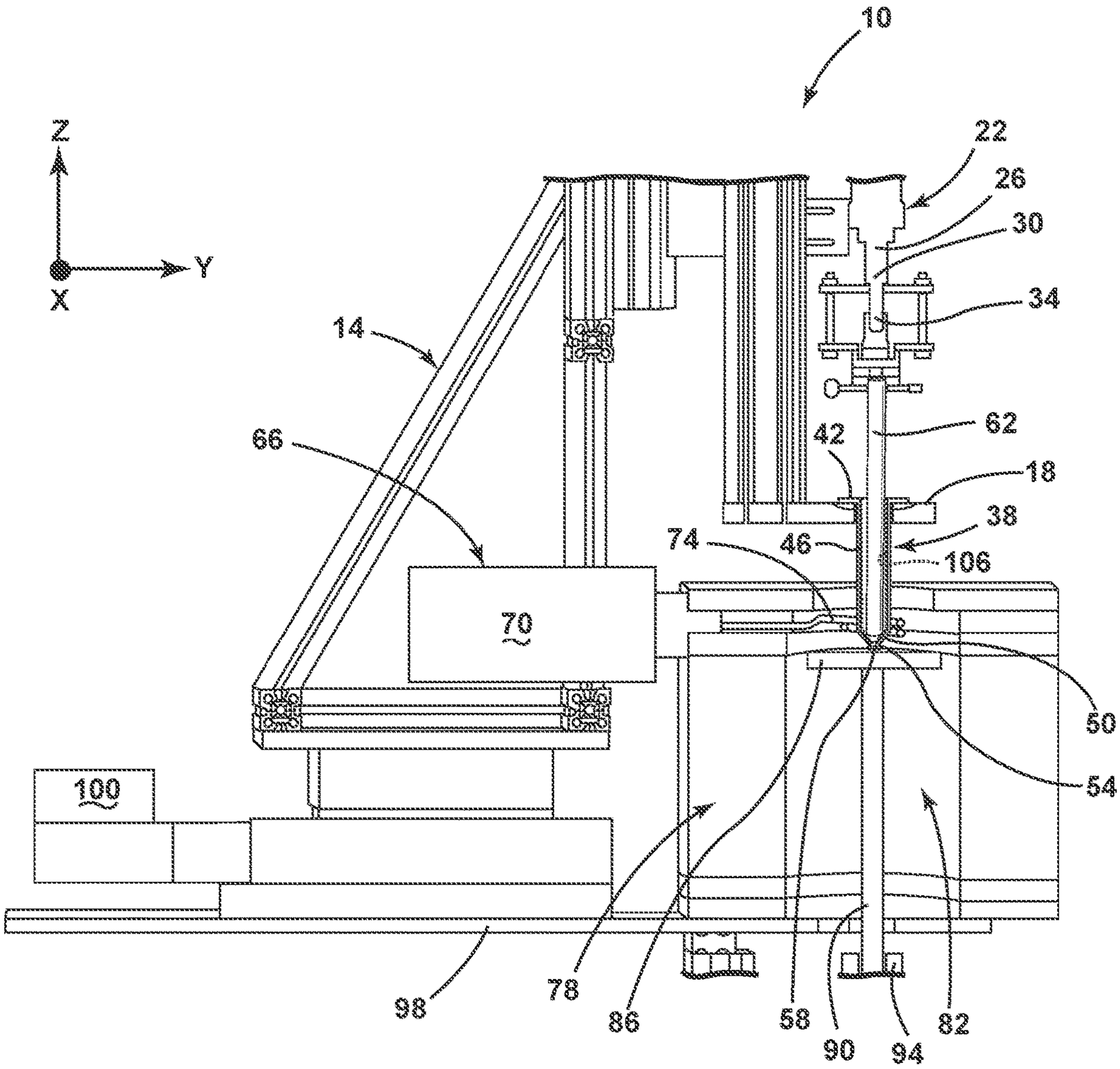

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FIG. 1A

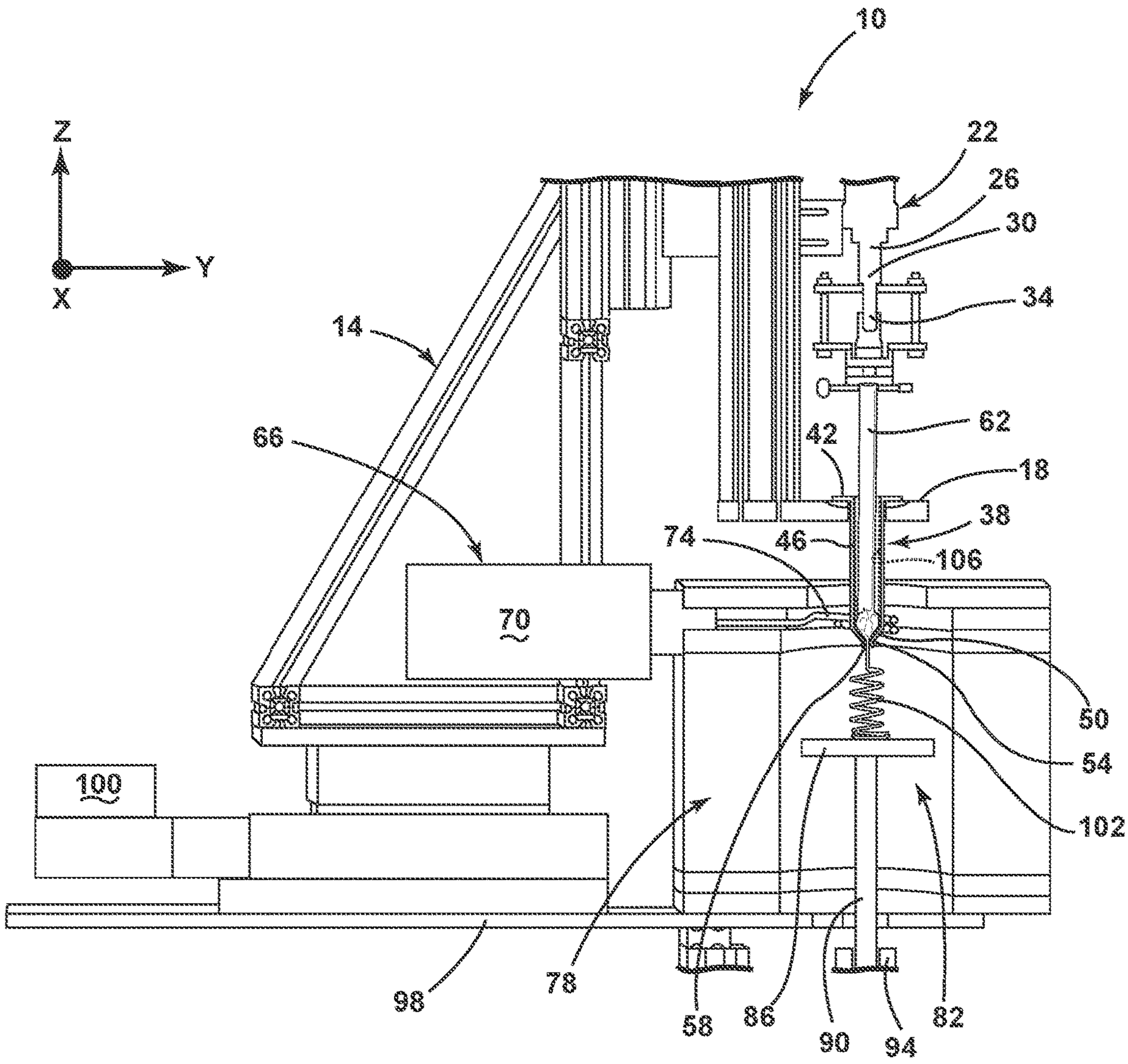

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FIG. 1B

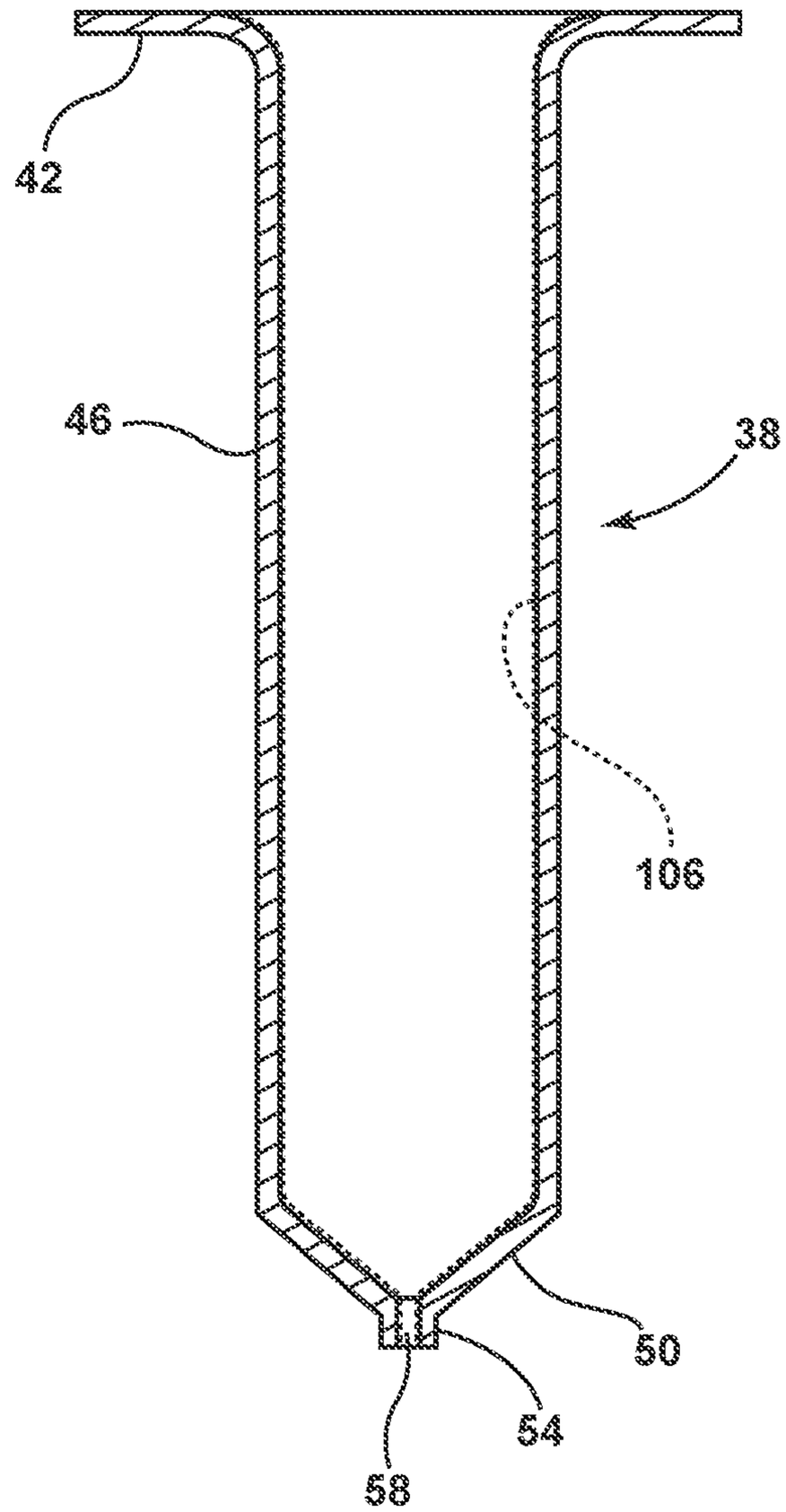


FIG. 2

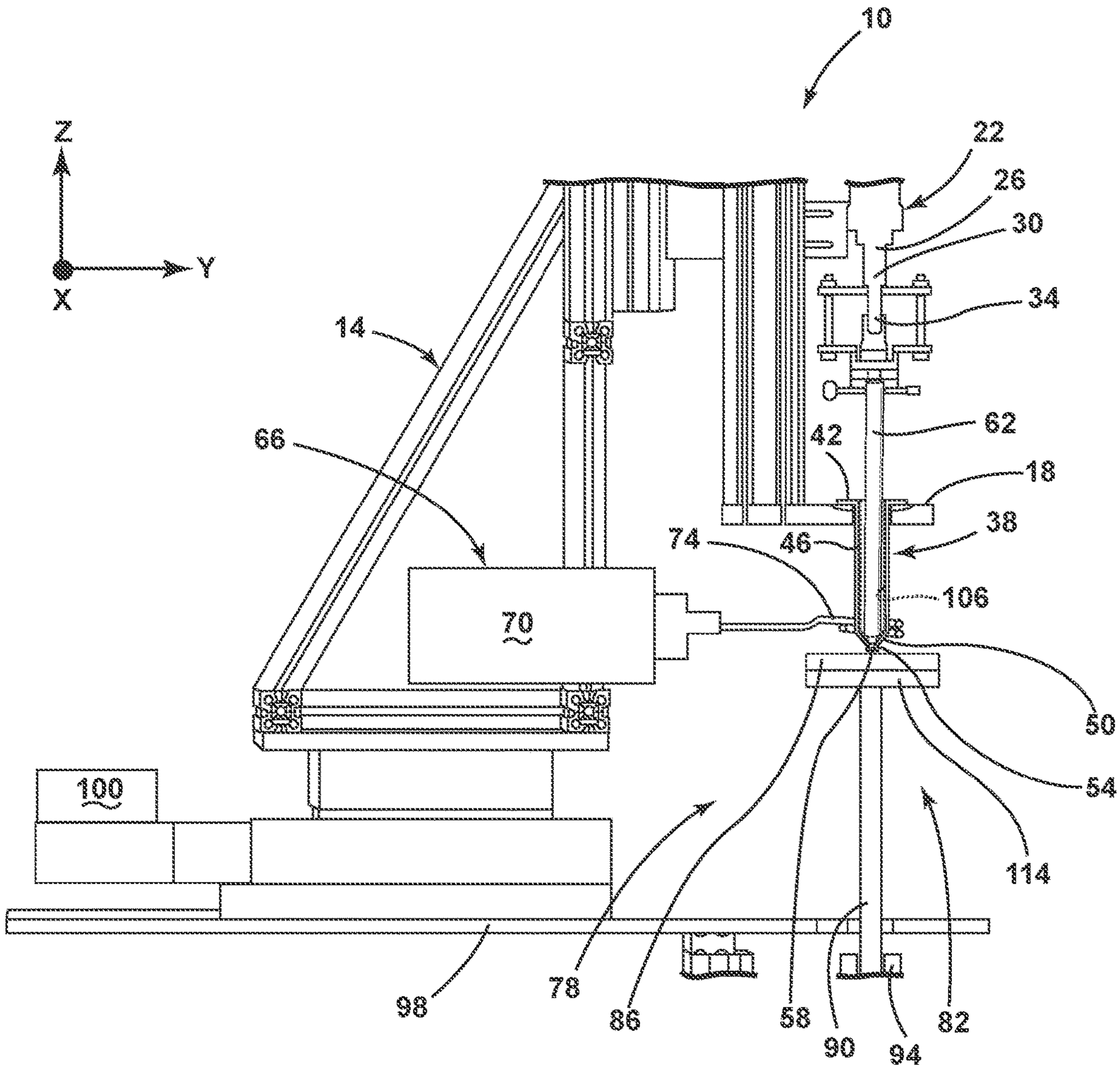

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78
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FIG. 3

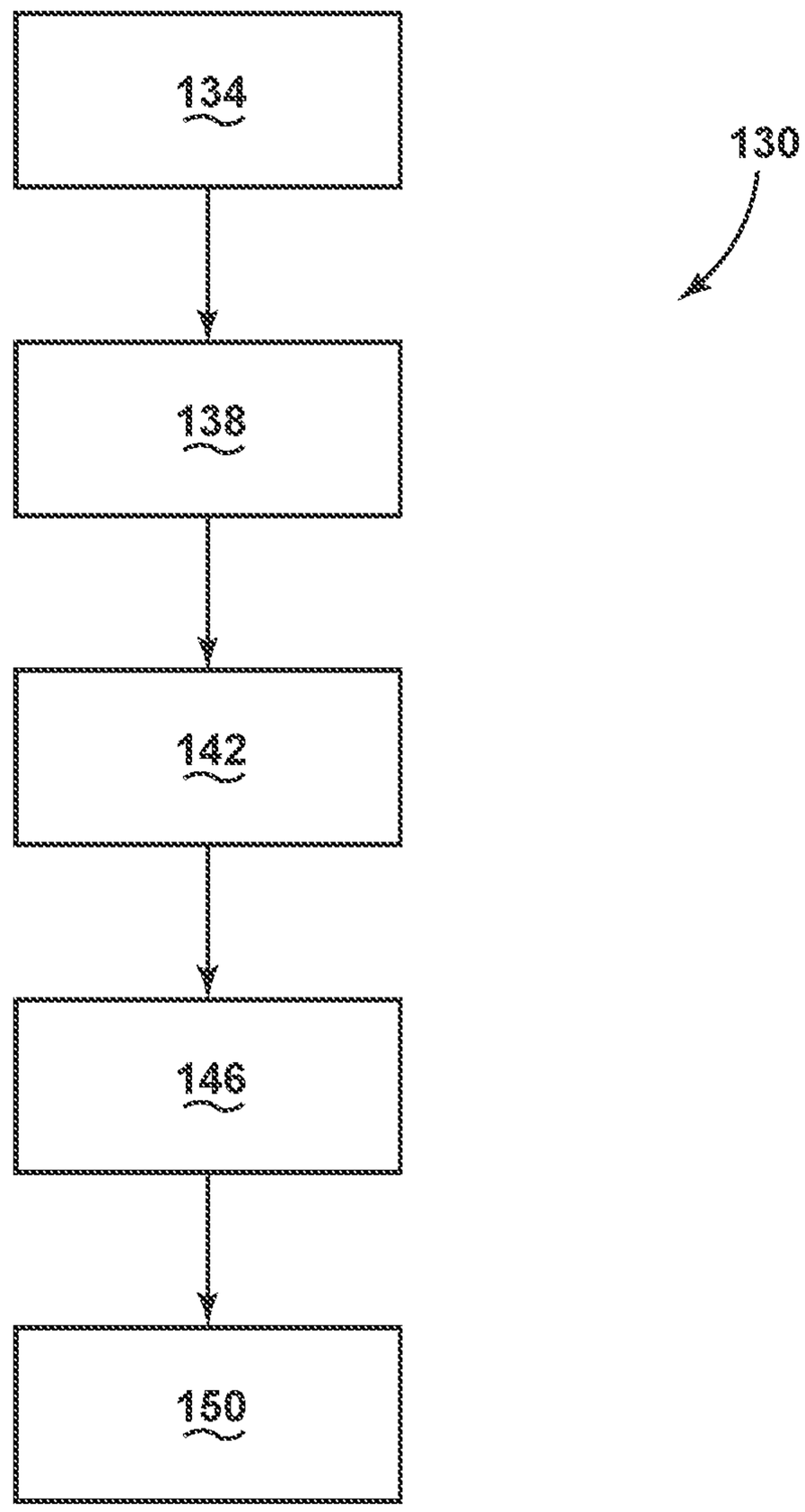


FIG. 4

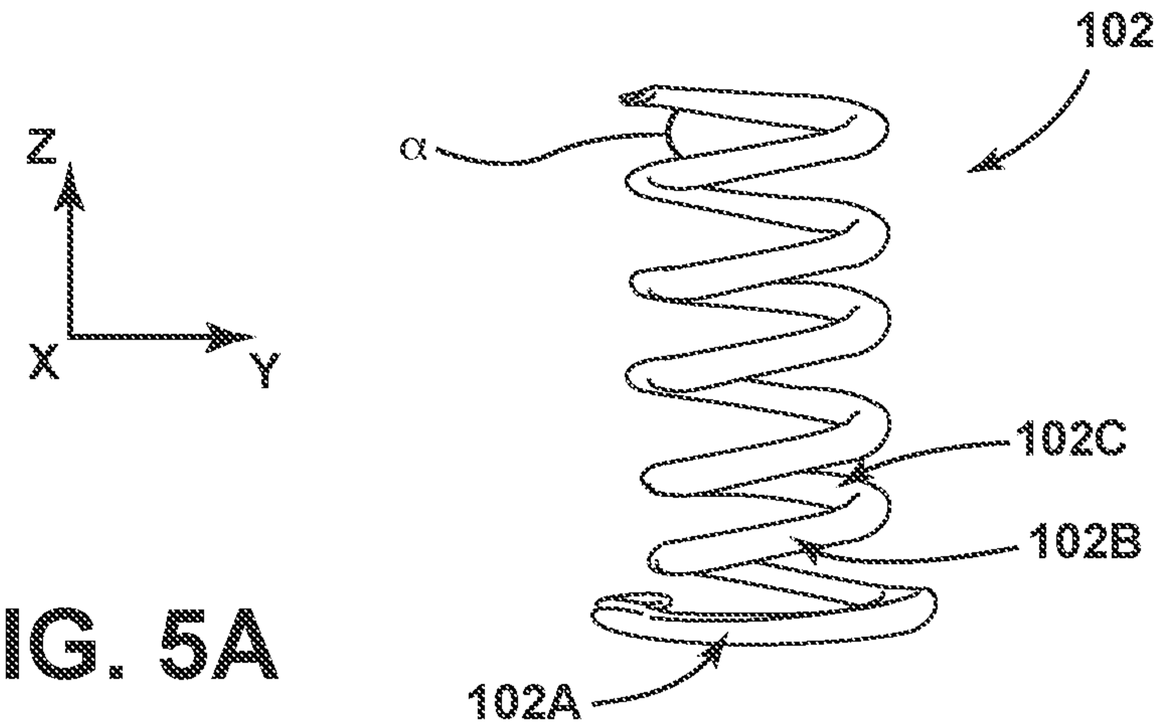


FIG. 5A

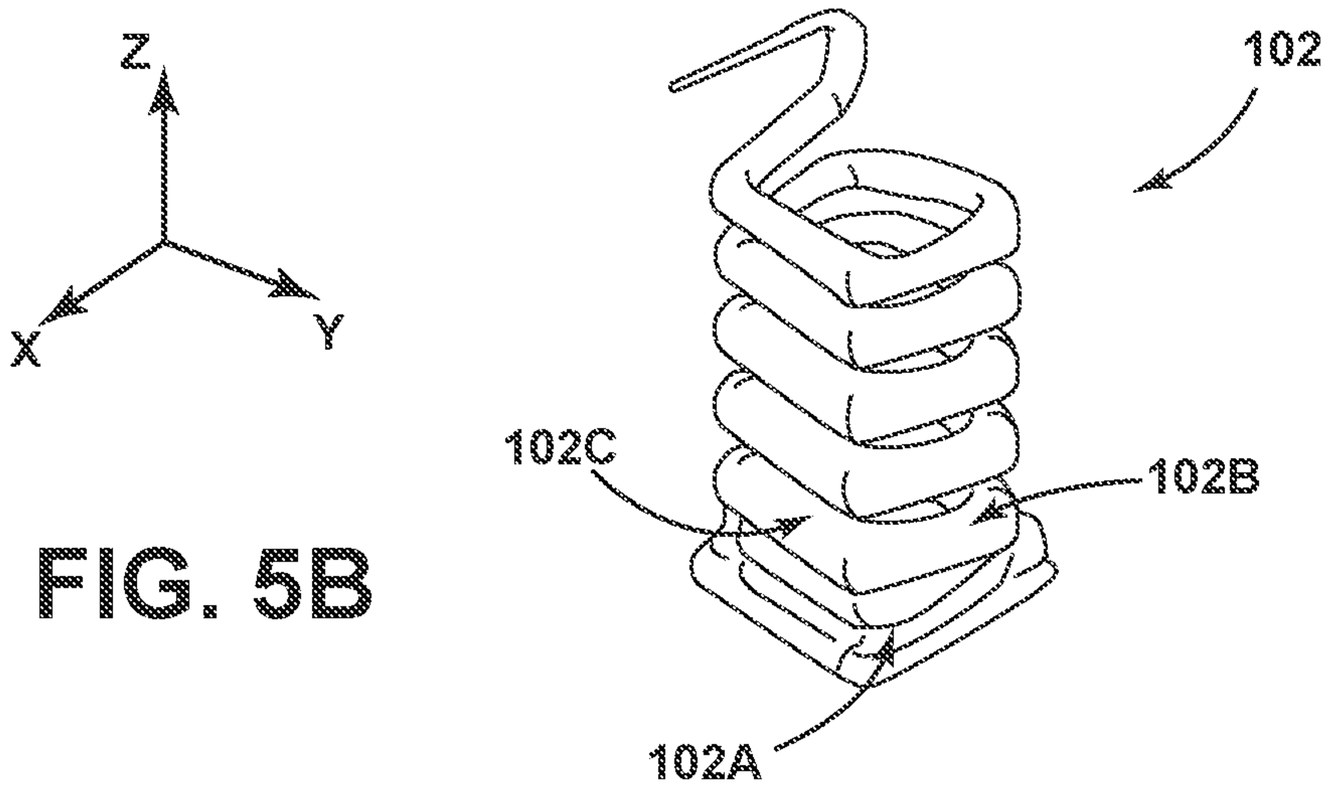


FIG. 5B

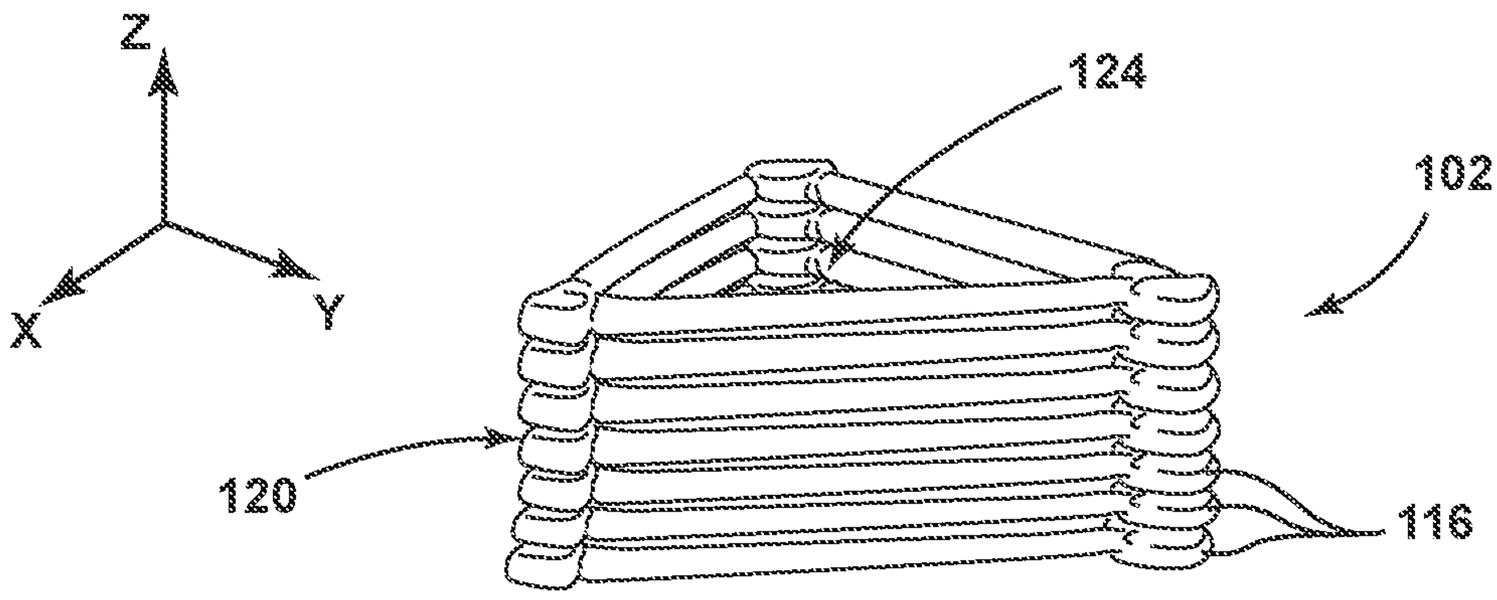


FIG. 5C

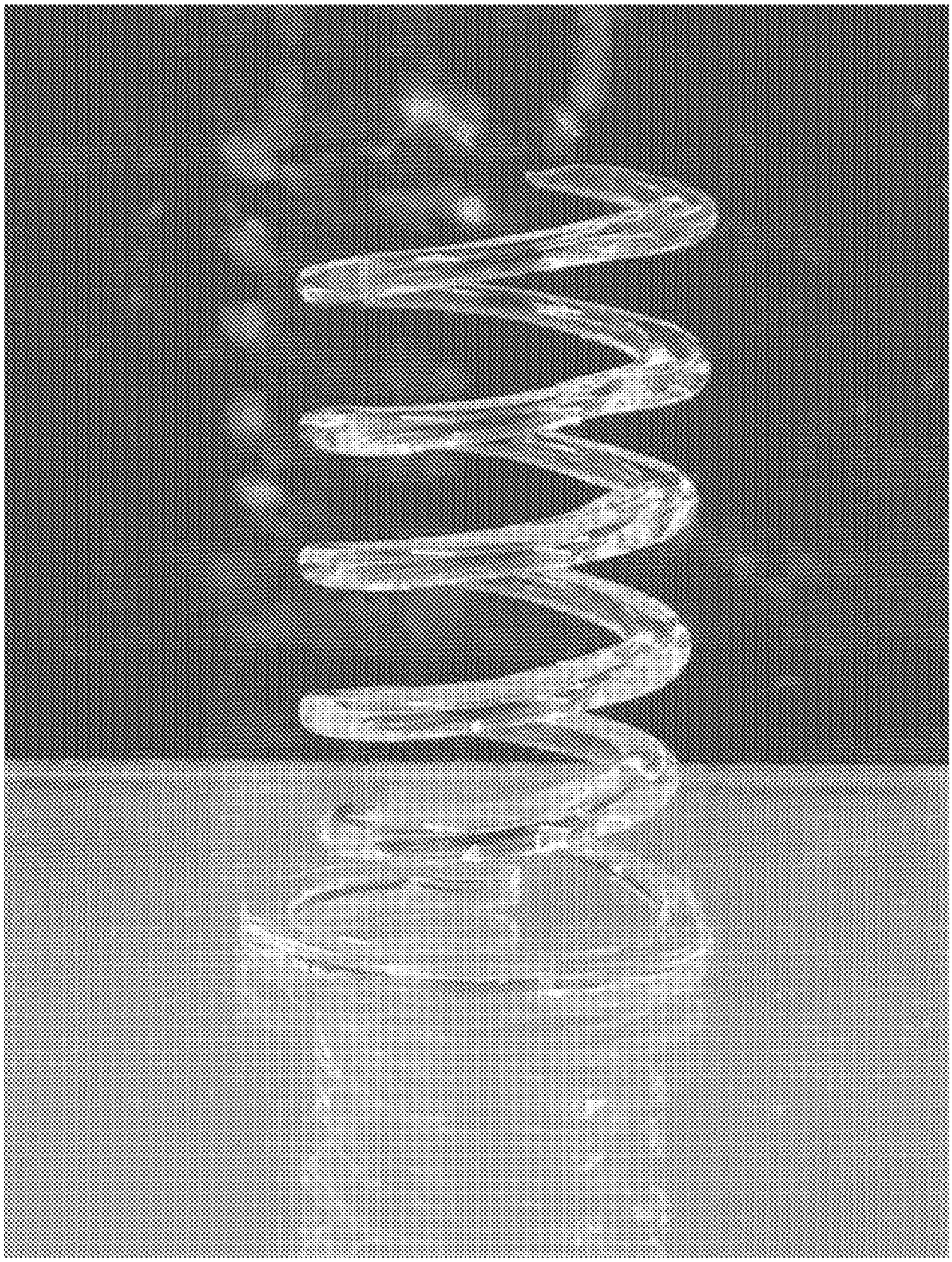


FIG. 6

ADDITIVE MANUFACTURING SYSTEMS AND METHOD FOR MAKING GLASS ARTICLES

[0001] This application claims the benefit of priority under 35 U.S.C. § 119 of U.S. Provisional Application Ser. No. 62/426,895, filed on Nov. 28, 2016, the contents of which are relied upon and incorporated herein by reference in their entirety

FIELD OF THE DISCLOSURE

[0002] The present disclosure generally relates to additive manufacturing systems, and more specifically, to an additive manufacturing system for forming glass articles.

BACKGROUND

[0003] Commonly available additive manufacturing techniques such as stereolithography of a resin filled with glass particles, or direct laser sintering of glass particles may have difficulty creating a part with excellent optical transparency because the glass particles may be difficult to sinter to full density. One additive manufacturing technique used for plastics, known as fused deposition modeling (FDM), has the advantage of using fiber as the feedstock, rather than a powder. In the FDM systems, fibers are pulled into a heated zone using a tractor wheel. Use of FDM with brittle glass fibers in place of the flexible plastic fibers results in broken fibers. In addition, it is not always possible to pull a fiber of the desired glass composition as the viscosity curve of flexible glass fibers is not always compatible with the fiber draw process. Conventional extrusion techniques may also be equally unsuited for additive manufacturing of glass products as extrusion is designed for larger diameters, and may require too high a temperature and pressure to produce a glass bead diameter of a desired size. Another method to lay down a thin bead of glass is to melt glass in a crucible with a hole at the bottom. However, as the diameter of the glass stream decreases, the stability of the stream decreases as well, and the flow stream may spiral and buckle.

SUMMARY OF THE DISCLOSURE

[0004] According to at least one aspect of the present disclosure, a glass article manufacturing system includes a crucible that defines a barrel and a nozzle. The barrel accepts a glass feedstock. A heater is in thermal communication with the nozzle. The heater heats the feedstock within the nozzle. An actuator is positioned proximate the barrel and extrudes the feedstock through the nozzle as extruded feedstock.

[0005] According to another aspect of the present disclosure, a glass article manufacturing system includes a crucible that defines a nozzle and accepts a glass feedstock. A platform is positioned proximate the nozzle. An actuator is positioned proximate the crucible and is arranged to apply pressure to the feedstock such that the feedstock extrudes through the nozzle onto the platform as an extruded glass feedstock. The extruded glass feedstock is in the form of a glass article.

[0006] According to another aspect of the present disclosure, a method of operating a glass article manufacturing system includes the steps: heating a glass feedstock within a crucible that defines a nozzle; extruding the glass feedstock through an aperture of the nozzle as a bead onto a platform; and moving the platform as the glass feedstock is extruded to form a glass article.

[0007] These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The following is a description of the figures in the accompanying drawings. The figures are not necessarily to scale, and certain features and certain views of the figures may be shown exaggerated in scale or in schematic in the interest of clarity and conciseness.

[0009] FIG. **1**A is a schematic diagram illustrating an additive manufacturing system at a start time, according to one embodiment;

[0010] FIG. **1**B is a schematic diagram illustrating an additive manufacturing system at an end time, according to one embodiment;

[0011] FIG. **2** is a schematic cross section of a crucible of the additive manufacturing system of FIG. **1**A, according to one embodiment;

[0012] FIG. **3** is a schematic diagram illustrating an additive manufacturing system, according to another embodiment;

[0013] FIG. **4** is a flow diagram of a method for operating the additive manufacturing system, according to one embodiment;

[0014] FIG. **5**A is top perspective view of a glass article formed using the additive manufacturing system, according to one embodiment;

[0015] FIG. **5**B is top perspective view of a glass article formed using the additive manufacturing system, according to one embodiment;

[0016] FIG. **5**C is a perspective view of a glass article formed using the additive manufacturing system, according to another embodiment; and

[0017] FIG. **6** is a photograph of an exemplary glass article formed by an additive manufacturing system, according to one embodiment.

DETAILED DESCRIPTION

[0018] Additional features and advantages of the invention will be set forth in the detailed description which follows and will be apparent to those skilled in the art from the description, or recognized by practicing the invention as described in the following description, together with the claims and appended drawings.

[0019] As used herein, the term "and/or," when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

[0020] In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions.

[0021] Referring to FIGS. **1**A-**3**, depicted is an additive manufacturing system **10** for making glass articles, among

other components. The system **10** includes a support structure **14** including an adapter **18**. In the depicted embodiment, an actuator **22** is positioned towards a top of the support structure **14**. The actuator **22** includes a servo **26**, a load cell **30** and a plunger **34**. Positioned below the actuator **22** is a crucible **38**. The crucible **38** includes a flange **42**, a barrel **46**, a knuckle **50**, a nozzle **54** and an aperture **58**.

[0022] In the embodiment depicted in FIGS. **1A-3**, the crucible **38** may be held to the support structure **14** by the adapter **18**. Positioned within the crucible **38** is a feedstock **62**. The system **10** further includes a heater **66**. The heater **66** includes an induction unit **70** and an induction coil **74**. A furnace **78** is positioned approximate the support structure **14**. The furnace **78** defines a cavity **82** into which the crucible **38** extends.

[0023] A platform **86** is positioned inside the cavity **82** of the furnace **78**. The platform **86** is supported by a support rod **90**. The support rod **90** is operably coupled to a Z-stage **94**. The Z-stage **94** is configured to move the platform **86** within the cavity **82** of the furnace **78** in a Z-direction. The support structure **14** is coupled to an XY-stage **98**. The Z-stage **94** and the XY-stage **98** are configured to move the platform **86** and the crucible **38** with respect to each other. It will be understood that the platform **86** and the furnace **78** may be arranged in a variety of configurations which allow movement relative to one another without departing from the teachings provided herein. For example, the platform **86** and/or furnace **78** may move circularly, cylindrically or in similar movements as defined by Cartesian or polar coordinates.

[0024] As will be explained in greater detail below, the additive manufacturing system **10** includes a controller **100** which is configured to regulate a pressure applied by the actuator **22**, the heat provided by the heater **66** to the crucible **38** (i.e., and the feedstock **62**), the movement of the platform **86** and the crucible **38** relative to each other, and the temperature of the furnace **78** to form a glass article **102**.

[0025] The support structure **14** is configured to hold various components of the system **10** in place during operation. The support structure **14** may include a linear slide to which the actuator **22** and/or the adapter **18** are coupled such that a crucible **38** and/or the actuator **22** may be adjusted in the Z-direction. The adapter **18** may include a groove to permit seating of the flange **42** of the crucible **38** to the adapter **18**. Insulators may be included on both sides of the flange **42** within the adapter **18** while ensuring proper seating of the crucible **38** within the support structure **14**. In some embodiments, these insulators may be washers or fiber blankets composed of a ceramic or polymeric material in order to provide electrical isolation to the crucible **38**. Further, the insulators may provide thermal insulation between the support structure **14** and the crucible **38**.

[0026] Positioned above the crucible **38** is the actuator **22**. It will be understood that the positional relationship between the actuator **22** and the crucible **38** may be changed depending on the glass article **102** intended to be made. For example, the crucible **38** and the actuator **22** may be positioned substantially at the same height such that the feedstock **62** is actuated in a substantially horizontal direction. The actuator **22** is configured to extend the plunger **34** in order to push the feedstock **62** toward the nozzle **54**. For example, the plunger **34** may be coupled in a gripping manner to the feedstock **62** to exert a downward force. In another example, the plunger **34** may press on a face of the feedstock **62** to force the feedstock **62** into the barrel **46** of the crucible **38**.

[0027] According to a specific embodiment, the servo **26** exerts a force on the plunger **34** which is then extended into the barrel **46**. The plunger **34** may have an outside diameter approximately equal to that of an inside diameter of the barrel **46**. In such examples, the plunger **34** may "wipe" an inner surface **106** of the barrel **46** such that all of the feedstock **62** is forced toward the nozzle **54**. In yet another example, the actuator **22** may include a roller for applying downward force to the feedstock **62**. The load cell **30** may measure the amount of force applied by the plunger **34**. The actuator **22** may provide from about 0.1 pounds (0.44 N) to about 300 pounds (1334 N) or more of force to the feedstock **62** within the crucible **38**. It will be understood that up to 1000 pounds (4448 N) of force may also be applied by the actuator **22** to the feedstock **62**. Further, the force applied to the feedstock **62** may be varied over time or though the formation of the glass article **102**.

[0028] According to various examples, the feedstock **62** may include one or more glasses and glass materials. The feedstock **62** may be formed as a rod having a diameter greater than or equal to about 1 mm, 20 mm, 30 mm, 40 mm, 50 mm, 100 mm, or larger than about 125 mm in diameter. A rod may be distinguished from a filament with respect to the thickness and the compressive force it may withstand as a rod is thicker than a filament and may withstand a greater compressive force. For example, while a filament may be flexible at room temperature, the rod example of the feedstock **62** may not be flexible at room temperature such that the force applied from the actuator **22** does not result in buckling or deformation of the feedstock **62**. It will be understood that the diameter of the rod of the feedstock **62** may be adjusted based on the desired size of the glass article **102** to be made. Further, the diameter of the feedstock **62** may be different over the length of the feedstock **62**. In other examples, the feedstock **62** may be composed of a plurality of rods (e.g., a bundle), a powder, a plurality of filaments, a plurality of disks (e.g., wafers or patties of the rods), a plurality of particles, a plurality of beads and/or combinations thereof.

[0029] As explained above, the feedstock **62** may be formed of a glass or glass material. The glass or glass material of the feedstock **62** may include Pyrex®, quartz, aluminum silicate glasses, soda-lime glass, an aluminosilicate glass, an alkali-aluminosilicate glass, a borosilicate glass, an alkali-borosilicate glass, an aluminoborosilicate glass, an alkali-aluminoborosilicate glass, a fused silica glass, glasses resistant to high thermal shock, glasses with high working ranges, colored glasses, doped glasses, transparent glasses, translucent glasses, opaque glasses and combinations thereof. It will be understood that the composition of the feedstock **62** may change or vary over the length of the feedstock **62**. For example, multiple different rods of different compositions of glass may be loaded into the crucible **38** such that at different points during extrusion of the feedstock **62** onto the platform **86**, different compositions of glass are formed. Such an embodiment may be advantageous in forming a glass article **102** having different regions of different composition.

[0030] According to various embodiments, the glass of the feedstock **62** may have a long working range. The working range of the glass is defined as the range of temperatures that

correspond to the point where the glass begins to soften to the point where the glass is too soft to control. In other words, the working range is the range of temperatures at which the viscosity of the feedstock **62** is sufficiently low enough to extrude, but not low enough as too melt and drip out of the nozzle **54**. Selection of the glass composition for the feedstock **62** is guided by choosing a glass with a viscosity curve, or working range, which does not result in a burdensome amount of temperature change to affect viscosity. Further, care should be taken during the selection of the glass composition to select a glass with a viscosity curve not so sensitive to temperature change that wide changes in viscosity occur over a short temperature range (e.g., less than 100° C., less than 50° C., less than 10° C.). In other words, when selecting a glass composition for the feedstock **62**, the composition should not be difficult to heat to a flowing state, but should also not be difficult to maintain in either a flowing state or a solid state. Glass compositions which include nodes in the viscosity change (i.e., drastic viscosity changes over a small temperature range) may be advantageous for various start and stop and sequences of the system **10**. The working range of the feedstock **62** may be greater than or equal to about 100° C., 150° C., 200° C., 275° C., 300° C., 350° C. or greater than about 500° C.

[0031] The crucible **38** holds the feedstock **62**. As explained above, the crucible **38** includes the flange **42**, the barrel **46**, the nozzle **54**, and defines the aperture **58**. The barrel **46** may have an inside diameter greater than or equal to about 10 mm, 20 mm, 30 mm, 34 mm, 40 mm, 50 mm, 100 mm, 200 mm or 500 mm. The barrel **46** may have a thickness of greater than or equal to about 1 mm, 2 mm, 5 mm, 10 mm, 25 mm or 50 mm. It will be understood that the thickness of the barrel **46** may be any practicable thickness for supporting the feedstock **62** under pressure from the actuator **22** and at temperature from the heater **66**. The aperture **58** may be positioned at the bottom of the crucible **38** such that the feedstock **62**, when heated (e.g., melted or otherwise heated to its working temperature), may be extruded therefrom. The aperture **58** may have an inside diameter of less than or equal to about 500 mm, 125 mm, 25 mm, 3 mm, 1.5 mm, 0.5 mm, or less than about 0.1 mm. It will be understood that the diameter of the aperture **58** may be altered depending on the size of the glass article **102** (e.g., larger aperture **58** for a larger glass article **102** to decrease manufacturing time) or based on a desired bead size of the feedstock **62** extruded through the aperture **58**.

[0032] The ratio between the inside diameter of the barrel **46** (e.g., an entrance to the nozzle **54**) and the aperture **58** may be greater than or equal to about 1, 1.5, 5, 10, 20 or 50. The nozzle **54** may define the aperture **58** as a variety of shapes including circular, square, triangular, star patterned, or other desired shapes of the bead of extruded feedstock **62**. Further, the nozzle **54** may be dynamic such that the size and/or shape of the aperture **58** may change throughout a process run of the system **10**. For example, the aperture **58** may begin at substantially circular, but may be changed to square or triangular part way through the process run and then optionally returned back to a circular shape. Further, the nozzle **54** may include a mandrel configured to extrude the feedstock **62** as a tube or other hollow structure. A plurality of thermocouples may be attached or otherwise coupled to the crucible **38** through the nozzle **54**, the knuckle **50** and the barrel **46** to measure the temperature of the feedstock **62** passing through the crucible **38** and different points.

[0033] The crucible **38** may be formed of a conductive metal such as platinum, rhodium, steel, stainless steel, and other metals with a melting temperature sufficiently above the working range of the feedstock **62**. In a specific example, the crucible **38** may be formed of an 80 weight percent (wt. %) platinum and 20 wt. % rhodium alloy. The crucible **38** may be formed of metal with a melting point greater than a softening point of the feedstock **62**. Metals of the crucible **38** may also be selected based on the reactivity of the metal with the glass. For example, metals which are not reactive with the feedstock **62** may be used. Reactivity between the feedstock **62** and the material of the crucible **38** may include the transfer of ions or elements between the feedstock **62** and the material of the crucible **38** to a point at which either the feedstock **62** and/or crucible **38** is unsuitable for its intended purpose (e.g., a property or characteristic changes).

[0034] Additionally or alternatively, the crucible **38** may include one or more inserts positioned between the barrel **46** and the feedstock **62**. The inserts may be formed of a different material than the crucible **38**. The inserts may take the form of a separate component inserted into the crucible **38** and/or take the form of a film or coating deposition on interior surfaces of the crucible **38**. Use of such inserts may be advantageous in broadening the materials that may be used for the crucible **38** (e.g., metals which otherwise be reactive with the feedstock **62**) by separating contact between the feedstock **62** and the material of the crucible **38**. For example, the crucible **38** can be made of stainless steel and the insert or film positioned on the inside of the crucible **38** may be a platinum rhodium alloy with low reactivity to the feedstock **62**. The metal selected for the crucible **38** may also be selected based on a creep resistance property. As the temperature of the crucible **38** increases, the force on the crucible **38** from the actuator **22** may result in a strain of the crucible **38**. Accordingly, materials having a high creep resistance, or low susceptibility to strain when under force at high temperatures, may be utilized for the crucible **38**.

[0035] According to various embodiments, at the beginning of a process run of the system **10**, the first rod of feedstock **62** inserted into the crucible **38** may be machined such that an exterior surface of the feedstock **62** substantially matches an interior surface of the nozzle **54** of the crucible **38** such that heat may be more efficiently transferred from the crucible **38** to the feedstock **62**. Such a machining of the feedstock **62** may lessen the amount of time necessary to begin producing the glass article **102**.

[0036] As explained above, the additive manufacturing system **10** includes the heater **66**. The heater **66** includes the induction unit **70** and the induction coil **74**. The induction unit **70** is configured to provide alternating current to the induction coil **74** such that the induction coil **74** may inductively heat the crucible **38**. In other words, the heater **66** is in thermal communication with nozzle **54** of the crucible **38**. The heat of the crucible **38** is then transferred to the feedstock **62** to heat the feedstock **62**. The amount of power provided by the induction unit **70** may be altered during a process run of the additive manufacturing **10** based on desired characteristics of the feedstock **62** as it is extruded into the glass article **102**. The induction coil **74** is depicted as surrounding the knuckle **50** of the crucible **38**, but it will be understood that the induction coil **74** may be positioned in a number of locations along the length of the crucible **38**. Further, multiple induction coils **74** may be utilized along the crucible **38** in order to heat various

locations of the feedstock **62**. Use of the induction coil **74** may be advantageous in providing nearly instantaneous control of the temperature of the crucible **38** and the feedstock **62**. It will be understood that the induction unit **70** and the induction coil **74** of the heater **66** may be replaced by other forms of heating the crucible **38**. For example, the heater **66** may be used in conjunction with, or replaced by, a flame heat system, an infrared heating system, a resistance coil heating system (e.g., a nichrome wrap) and other forms of heating.

[0037] In the depicted embodiment, the furnace **78** is positioned below the crucible **38**. The crucible **38** extends into the cavity **82** of the furnace **78**. It will be understood that the crucible **38** may extend into the furnace **78** or the aperture **58** may be coplanar with an entrance of the furnace **78**. The furnace **78** may be sealed at a top and a bottom to keep a heated environment within the furnace **78**. The cavity **82** of the furnace **78** may be filled with an inert gas (e.g., non-reactive to the glass article **102** over the feedstock **62**) or may be filled with typical atmospheric gases. The furnace **78** may keep a temperature sufficiently high to anneal the glass article **102** but lower than the working temperature of the feedstock **62**. The temperature of the furnace **78** may be sufficiently high to keep the extruded glass article **102** pliable, but not high enough to allow sag in the article **102**.

[0038] The platform **86** is positioned within the cavity **82** of the furnace **78**. It will be understood that the platform **86** may be replaced with any build surface or substrate. As explained above, the platform **86** is positioned within the furnace **78** to accept or receive the extruded glass feedstock **62**. It will be understood that a component (e.g., a mechanical and/or electrical part) may be placed on the platform **86** and received the feedstock **62** such that the glass article **102** is a subcomponent of a larger component. The support rod **90** extends from a bottom of the platform **86**, through the cavity **82** and out of the furnace **78**. The support rod **90** is coupled with the Z-stage **94** such that the platform **86** may be raised and lowered in the Z-direction. Further, the support structure **14** is coupled with the XY-stage **98** such that the nozzle **54** and the platform **86** may be moved in the X-, Y- and Z-directions relative to each other. According to at least one alternative example, the support structure **14** may be coupled to the Z-stage **94** and the XY-stage **98** such that the controller **100** may regulate movement of the crucible **38** relative to the platform **86**. Such an example may be advantageous for the production of large glass articles **102** (i.e., such that the large glass article **102** does not have to be moved). In another alternative example, the platform **86** may be coupled to the Z-stage **94** and the XY-stage **98** such that the controller **100** may regulate movement of the platform **86** relative to the crucible **38**. Such an example may be advantageous for the production of smaller glass articles **102** (i.e., because the relatively larger support structure **104** may remain stationary). Even further, all or some of the system **10** may be positioned within the furnace **78** for the production of large glass articles **102**.

[0039] According to some embodiments, a heating element **114** (FIG. **3**) may be positioned on a bottom of the platform **86**. The heating element **114** may extend over all or a portion of the platform **86**. The heating element **114** may be configured to heat all of or just a portion of the platform **86** (i.e., to form hot and cold zones on the platform **86**). As such, the platform **86** may form a heated build surface. Such hot and cold zones may be advantageous in manufacturing the glass article **102** to have different properties throughout its structure. Heating of the platform **86** by the heating element **114** may decrease a thermal shock experience by the glass article **102** as the feedstock **62** is extruded from the crucible **38**. Use of the heating element **114** may be advantageous in embodiments of the additive manufacturing system **10** not incorporating the furnace **78** (e.g., FIG. **3**) or an embodiments where the furnace **78** is kept at a lower temperature. It will be understood that in commercial examples of the system **10**, the platform **86** may be a portion of a conveyor belt or other assembly line component configured to mass produce the glass articles **102**. In such an example, the crucible **38** may be configured to move relative to the platform **86**.

[0040] In operation of the system **10**, the controller **100** is configured to instruct the actuator **22** to exert a force on the feedstock **62** to move the feedstock **62** into the crucible **38**. As the crucible **38** is heated, the heat is transferred to the feedstock **62**. The feedstock **62** is heated to a temperature within its working range such that the feedstock may begin to flow through the aperture **58** of the nozzle **54** under the pressure from the actuator **22**. As such, the feedstock **62** is extruded through the nozzle **54** of the crucible **38**. The feedstock **62** may be heated proximate the knuckle **50** and the nozzle **54**, but also at points throughout the barrel **46**. The feedstock **62** exits the nozzle **54** as a continuous bead of material. The feedstock **62** then contacts the platform **86** and begins to "set up," or cool as it is extruded. In other words, as the feedstock **62** contacts the platform **86**, the feedstock **62** cools and increases in viscosity until the feedstock **62** solidifies.

[0041] After the bead of feedstock **62** contacts the platform **86**, the platform **86** may begin to move in a 3-dimensional manner using the Z-stage **94** and/or the XY-stage **98**. As explained above, additionally or alternatively, the crucible **38** may be moved relative to the platform **86** (e.g., for the production of large glass articles **102**). As the platform **86** is moved relative to the nozzle **54**, the bead of feedstock **62** begins to extend through space (i.e., and solidify as it goes) to form the glass article **102**. In other words, the feedstock **62** solidifies as it is extruded such that the glass article **102** maintains the shape generated by the relative motion of the platform **86** and the nozzle **54**. At an end point of the glass article **102**, the controller **100** controls the heater **66** to stop heating of the crucible **38** which in turn returns the feedstock **62** to a temperature lower than its working range. The relatively quick reduction of the temperature of the feedstock **62** and crucible **38**, in addition to a removal of the force applied by the actuator **22**, causes the feedstock **62** to suck back into the nozzle **54** due to a negative pressure. Further, the actuator **22** may pull back on the feedstock **62** resulting in the feedstock **62** being sucked back into the nozzle **54**. Such a quick temperature shift and recoiling of the feedstock **62** back into the nozzle **54** may help starting and stopping the material flow, and reducing or eliminating "hairs," or fine strands of material extending away from the glass article **102** toward the nozzle **54**, at the article's end point. Further, a rapid motion by the nozzle **54** at the end of the run (relative to the formed glass article's end point), in addition to the change in temperature and pressure, may remove hairs from an end point of the glass article **102**. The controller **100** may control the actuator **22** and platform **86** in concert to create the glass article **102** from a single continuous bead of feedstock **62**, from a plurality of beads

of feedstock **62** laid on one another, or combinations thereof. At hotter temperatures of extrusion and/or of the furnace **78**, the beads of feedstock **62** may merge into a seamless, optically transparent, multilayer structure.

[0042] Referring now to FIG. **4**, depicted is an exemplary method **130** of operating the additive manufacturing system **10** to produce the glass article **102** (FIG. **1**A). The method **130** begins with step **134** of inserting the feedstock **62** into the crucible **38** if the system **10**. The feedstock **62** may be coupled to the actuator **22** at the same time. Next, step **138** of heating the glass feedstock **62** within the crucible **38** is performed. As explained above, the heater **66** heats the crucible **38** which in turn heats the glass feedstock **62** within the crucible **38**. The heater **66** heats the feedstock **62** to a sufficiently high temperature such that the feedstock **62** is within its working range.

[0043] Next, step **142** of extruding the glass feedstock **62** through the nozzle **54** onto the platform **86** is performed. In step **142**, the actuator **22** applies sufficient force to the feedstock **62** such that the portion of the feedstock **62** heated to its working range is extruded through the nozzle **54** and onto the platform **86**. The feedstock **62** is extruded as a bead. The controller **100** may control the actuator **22** to extrude a single, continuous, bead or a plurality of smaller beads of feedstock.

[0044] Next, step **146** of moving at least one of the crucible **38** and the platform **86** is performed. As explained above, the controller **100** is configured to regulate positional control of the crucible **38** and/or the platform **86** relative to one another. The controller **100** is configured to move the crucible **38** and/or the platform **86** as the feedstock is extruded from the nozzle **54** to form the glass article **102**. The controller **100** controls the position of the crucible **38** and/or platform **86** such that the bead(s) of feedstock **62** is placed on the platform to build the glass article **102**. While moving the crucible **38** and/or the platform **86**, the controller **100** may be configured to drag the nozzle **54** through the previously applied bead of feedstock **62**. The nozzle **54** may be dragged through the bead at a depth less than or equal to about half the thickness of the material layer being deposited. Dragging the nozzle **54** through the bead of feedstock **62** on the platform **86** may be advantageous in helping to smear the previously laid bead of feedstock **62** and create better adhesion between beads of feedstock **62** laid on top of one another. Better adhesion between the beads may result in tighter stack-up tolerances.

[0045] Next, step **150** of annealing the glass article **102** may be performed. Annealing of the glass article **102** may be performed in the furnace **78** and the temperature and time at which the glass article **102** is annealed may be regulated by the controller **100**.

[0046] It will be understood that the steps of the method **130** may be performed in any order, repeated, omitted and/or performed simultaneously without departing from the teachings provided herein.

[0047] Referring now to FIGS. **5**A-**5**C, depicted are various embodiments of the glass article **102** as manufactured by the system **10**. According to various examples, the glass article **102** may be substantially transparent and/or colorless. The glass article **102** may have a transparency greater than about 60%, 70%, 80%, 90% or greater than about 99% for visible light. The glass article **102** is composed of one or more beads extruded proximate one another to form the glass article **102**. For example, the glass article **102** may include a single bead (FIGS. **5**A and **5**B) extending through a three dimensional space or a single or multiple beads stacked on one another (e.g., FIG. **5**C).

[0048] In single bead examples, the glass article **102** may define a base portion **102**A, a first body portion **102**B and a second body portion **102**C. The first and second body portions **102**B, **102**C may be coupled such that a self-supporting angle α between the first and second body portions **102**B, **102**C is less than or equal to about 45°. The glass articles **102** may have a self-supporting angle α of less than about 45°, 30°, 20°, 10° or less than about 1° as measured in an XZ and/or YZ plane relative to a horizontal XY plane. It will be understood that the self-supporting angle α may be formed at any angle between about 0.1° and about 180°. For purposes of this disclosure, the self-supporting angle α is the angle at which the glass article **102** may support an extension without an additional support structure (e.g., a tower or additional mold piece configured to hold up the extension of the glass article **102**). In other words, the self-supporting angle α has no supporting structure that extends between the first and second body portions **102**B, **102**C. Conventional additive manufacturing systems often utilize one or more fugitive materials to form a support structure. The fugitive material may be etched, melted and/or burned away after formation of the article to form the self-supporting angle α. The presently disclosed system **10** may be capable of forming the self-supporting angle α in the glass article **102** without the use of fugitive materials and/or a support structure. It is believed that such self-supporting angles α are feasible because the glass feedstock **62** sets up as it is extruded onto the platform **86**. In other words, it is believed that the feedstock **62** sufficiently solidifies as it is extruded to provide enough strength to form the self-supporting angle α. Such self-supporting angles α allow considerable over-hang as compared to articles formed using conventional additive manufacturing systems. Further, the glass article **102** may exhibit bends, or changes of direction, of less than about 135°, 90°, 45°, 10° or less than about 1°. It will be understood that a bend or change in direction of the glass article **102** may be between about 0.1° and about 359°.

[0049] In alternative examples, the glass article **102** may be formed of a plurality of glass beads arranged in a stack to form the three-dimensional glass article **102**. In such an example, each bead may be fused to an adjacent bead. It will be understood that although described as a plurality of beads, the glass article **102** may be formed from a single continuous bead folded or guided back onto its self. The beads may be fused to one another over the length of the beads or at a plurality of points. In such examples, the glass article **102** may be substantially transparent through the stack of fused beads. As explained above, the beads of extruded feedstock **62** may flow into crevices formed between adjacent beads which may enhance the transparency of the glass article **102** (e.g., due to elimination of air voids between the beads). Further, the glass article **102** may define one or more voids within the article **102** formed through placement of the beads of feedstock **62**. As explained above, by positioning, or dragging, the nozzle **54** in a previously laid bead of the feedstock **62**, the stack-up tolerance of the glass article **102** may be minimized with respect to conventional glass additive manufacturing techniques. The glass article **102** may take a variety of configurations. For example, the glass article **102** may form a glass encapsulation device (e.g., for electronic devices), a flow

reactor, or a nose cone with conformal cooling channels. The glass article **102** may be substantially or completely bubble free and may be of a complex design. As explained above, the composition of the glass article **102** may vary across the stack (i.e., in multiple bead or stacked single bead examples) and/or across individual beads.

[0050] A variety of advantages may be obtained using the disclosure provided herein. First, the additive manufacturing system **10** may produce a glass article **102** which is substantially transparent, bubble free and of a complex design. Second, the glass article **102** may have an increased overhang with the respect to conventional additive manufacturing techniques due to the decrease in self-supporting angle α provided by the system **10**. Third, use of the furnace **78** may prevent a thermally induced curl in the glass article **102** and may prevent the glass article **102** from undergoing a thermal shock. Fourth, complex designs, including tubes, may be formed in the glass article **102**. Fifth, the improved starts/stop control of the system **10** results in increased consistency at an end point of the glass article **102** (e.g., a decrease in the production of "hairs"). A decrease in the presence of hairs may allow for a more aesthetically pleasing and complex article **102** to be formed. Sixth, the system **10** may extrude a bead of the feedstock **62** onto an existing component to form a glass portion of that component. Seventh, the composition and/or properties (e.g., color, transparency, resistance to thermal shock, etc.) of the feedstock **62** may be altered through the process run that different portions of the glass article **102** exhibit different properties. Eighth, as the feedstock **62** is extruded and solidifies, molds and other conventional forming techniques for glass components may not be necessary which may save manufacturing time and cost. Ninth, the system **10** is scalable to produce glass articles **102** of nearly any size by changing the size of the crucible **38**, nozzle **54** and/or actuator **22**. Tenth, use of the rod examples of the feedstock **62** instead of traditional filaments allows longer operating times between when the system **10** must be reloaded with more feedstock **62**.

Example

[0051] Depicted in FIG. **6** is a photograph of a glass structure (e.g., the glass article **102**) produced using a three dimensional glass printer (e.g., the system **10**). As can be seen, the glass structure is substantially transparent and exhibits a substantial overhang due to the structure's low self-supporting angle (e.g., less than about 45°). The structure is formed from a single, continuous, bead of glass through three dimensional space. The bead exhibits a smooth upward curve to provide a general "cork screw" form to the glass structure. A feed material (e.g., feedstock **62**) used by the printer was Pyrex® glass.

[0052] Modifications of the disclosure will occur to those skilled in the art and to those who make or use the disclosure. For example, the plunger **34** of the actuator **22** may be replaced by rollers configured to exert a downward force on the feedstock **62**. In another example, the system **10** may be used to form glass articles **102** having a simple, substantially two dimensional, shape. Therefore, it is understood that the embodiments shown in the drawings and described above are merely for illustrative purposes and not intended to limit the scope of the disclosure, which is defined by the following claims, as interpreted according to the principles of patent law, including the doctrine of equivalents.

[0053] For purposes of this disclosure, the term "coupled" (in all of its forms: couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature, or may be removable or releasable in nature, unless otherwise stated.

1. A glass article manufacturing system, comprising:
a crucible comprising a barrel and a nozzle, the barrel accepts a glass feedstock;
a heater in thermal communication with the nozzle, the heater heats the feedstock within the nozzle;
a furnace positioned proximate the nozzle that anneals the extruded feedstock; and
an actuator positioned proximate the barrel that extrudes the feedstock through the nozzle as extruded feedstock.

2. The system of claim **1**, further comprising:
a build surface that receives the extruded feedstock from the nozzle.

3. The system of claim **2**, wherein the build surface comprises a heated surface proximate the nozzle.

4. (canceled)

5. The system of claim **2**, further comprising:
a controller that regulates movement of the crucible and the build surface relative to each other.

6. The system of claim **1**, wherein the heater comprises an induction coil, a resistance coil or combinations thereof.

7. The system of claim **1**, wherein the crucible comprises a metal with a melting point greater than a softening point of the feedstock.

8. The system of claim **1**, wherein the actuator comprises a plunger that presses the feedstock.

9. The system of claim **8**, wherein the plunger is further arranged to wipe an inside surface of the barrel.

10. The system of claim **1**, wherein the glass feedstock is a rod having a diameter greater than about 1 mm.

11. The system of claim **1**, wherein a composition of the glass feedstock varies over a length of the feedstock.

12. The system of claim **1**, further comprising:
an insert positioned between the barrel and the feedstock.

13. A glass article manufacturing system, comprising:
a crucible comprising a nozzle, the crucible accepts a glass feedstock;
a platform positioned proximate the nozzle; and
an actuator positioned proximate the crucible and arranged to apply pressure to the feedstock such that the feedstock extrudes through the nozzle onto the platform as an extruded glass feedstock, the extruded glass feedstock in the form of a glass article.

14. The system of claim **13**, wherein the extruded glass feedstock is substantially transparent and has a working range of greater than or equal to 100° C.

15. The system of claim **13**, further comprising:
a controller to regulate a heater in thermal communication with the nozzle.

16. The system of claim **15**, wherein the controller regulates a furnace positioned proximate the nozzle to anneal the glass article.

17. The system of claim **15**, wherein the heater comprises an induction coil, a resistance coil, or combinations thereof.

18. The system of claim **13**, further comprising:
a heating element to heat the platform.

19. The system of claim **15**, wherein the controller regulates movement of the platform relative to the crucible.

20. The system of claim **15**, wherein the controller regulates movement of the crucible relative to the platform.

21. The system of claim **15**, wherein the controller regulates movement of the platform and the nozzle in X-, Y- and Z-directions relative to one another.

22. A method of operating a glass article manufacturing system, comprising the steps:
heating a glass feedstock within a crucible comprising a nozzle;
extruding the glass feedstock through an aperture of the nozzle as a bead onto a platform; and
moving the platform as the glass feedstock is extruded to form a glass article.

23. The method of claim **22**, further comprising the step:
annealing the glass article.

24. The method of claim **22**, wherein the glass feedstock is a rod.

25. The method of claim **22**, further comprising the step of:
dragging the nozzle through the bead.

26. The method of claim **22**, wherein the platform is heated.

27. A glass article formed by the system of claim **1**, comprising:
a base portion;
a first body portion coupled to the base portion; and
a second body portion coupled to the first base portion, the first and second body portions coupled at a self-supporting angle of less than about 45° relative to an XZ or YZ plane.

28. The glass article of claim **27**, wherein the self-supporting angle is less than about 40°.

29. The glass article of claim **27**, wherein no supporting structure extends between the first and second body portions.

30. The glass article of claim **27**, wherein the glass article is substantially transparent.

31. The glass article of claim **27**, wherein the base portion, the first body portion and the second body portion are integrally defined.

32. The glass article of claim **27**, wherein a composition of the glass article varies throughout the glass article.

33. A glass article formed by the method of claim **22**, comprising:
a plurality of glass beads arranged in a stack to form a three-dimensional object, each bead being fused to an adjacent bead, wherein the article is substantially transparent through the fused beads.

34. The glass article of claim **33**, wherein the stack of glass beads defines a bend of less than about 90°.

35. The glass article of claim **33**, wherein the stack defines a void within the glass article.

36. The glass article of claim **33**, wherein the stack defines a self-supporting angle in a XZ or YZ plane of less than or equal to about 45° between adjacent beads.

37. The glass article of claim **33**, wherein a composition of the glass article varies across the stack.

38. The glass article of any of claim **33**, wherein a composition of the glass article varies across at least one bead.

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